F79MA - Statistical Models A

Final Exam Solution (December)

1. (a) i.

$$f(x;\theta) = \begin{cases} e^{(\theta - x)} & , \ x > \theta \\ 0 & , \ \text{otherwise} \end{cases}$$

$$f(x;\theta) = e^{(\theta-x)}\mathbb{I}_{(\theta,\infty)}(x)$$

$$\begin{aligned} L(\theta) &= \prod_{i=1}^{n} f(x_i,\theta) \\ &= \prod_{i=1}^{n} e^{(\theta - x_i)}\mathbb{I}_{(\theta,\infty)}(x) \\ &= e^{(n\theta - \sum_i^n x_i)} \prod_{i=1}^{n} \mathbb{I}_{(\theta,\infty)}(x) \\ &= e^{(n\theta - \sum_i^n x_i)}\mathbb{I}_{(\theta,\infty)}(x_{(1)}), \\ \text{where } x_{(1)} &= \min(x_1, x_2, \ldots, x_n). \end{aligned}$$

$$\therefore f(x;\theta) = \underbrace{e^{-\sum_i^n x_i}}_{h_1(\underline{x})} \underbrace{e^{n\theta}\mathbb{I}_{(\theta,\infty)}(x_{(1)})}_{h_2(\theta, x_{(1)})}$$

Hence by the factorisation theorem, $X_{(1)}$ the smallest order statistic is the sufficient statistic.

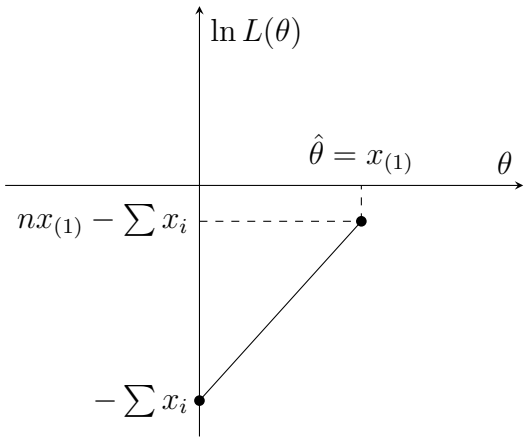
ii.

$$\begin{aligned} \mathrm{L}(\theta) &= \prod_{i=1}^{n} f(x_i, \theta) \\ &= \prod_{i=1}^{n} e^{(\theta - x_i)} \quad (\text{all } x_i > \theta) \\ &= e^{n\theta - \sum x_i}. \end{aligned}$$

$$\therefore \ln \mathrm{L}(\theta) = n\theta - \sum_{i=1}^{n} x_i$$

Hence the log likelihood function, $\ln \mathrm{L}(\theta)$ is a straight line funtion of θ which is graphed below.

Graph of $\ln L(\theta)$



Clearly it can be seen from the graph, that the maximum of $\mathrm{L}(\theta)$ occurs at $\hat{\theta} = X_{(1)}$ (i.e. the smallest order statistic).

(b)

$$f(y;\theta) = (\ \exp(-\theta)\ )^{1-y}(\ 1-\exp(-\theta)\ )^{y}, \quad y = 0, 1.$$

Therefore,

$$\mathbb{P}(Y=0) = \exp(-\theta) \text{ and } \mathbb{P}(Y=1) = 1-\exp(-\theta).$$

$$\begin{aligned}\mathbb{E}[Y] &= 0 \cdot \mathbb{P}(Y=0) + 1 \cdot \mathbb{P}(Y=1) \\ &= \mathbb{P}(Y=1) \\ &= 1-\exp(-\theta).\end{aligned}$$

Equate sample moments to population moment, $\overline{Y} = \mathbb{E}[Y]$.

$$\begin{aligned}\implies \overline{Y} &= 1-\exp(-\theta) \\ \exp(-\theta) &= 1-\overline{Y} \\ \frac{1}{\exp(\theta)} &= 1-\overline{Y} \\ \exp(\theta) &= \frac{1}{1-\overline{Y}} \\ \theta &= \ln\left(\frac{1}{1-\overline{Y}}\right) \\ &= -\ln(1-\overline{Y})\end{aligned}$$

Hence, the moments estimator is given by,

$$\underline{\underline{\tilde{\theta} = -\ln(1-\overline{Y})}}$$

Q2: (i) (Similar to tutorial)

$$L(\alpha; \underline{X}) = \frac{\alpha^n}{\theta^n} \prod_{i=1}^{n} \left(1 + \frac{x_i}{\theta}\right)^{-\alpha-1} \quad (+2)$$

ii) Similar to tutorial, different dist

$$\log L(\alpha; \underline{X}) = n \log(\alpha) - \alpha \sum_{i=1}^{n} \log\left(1 + \frac{X_i}{\theta}\right) + \dots \quad (+1)$$

$$\frac{\partial}{\partial \alpha} \log L = \frac{n}{\alpha} - \sum_{i=1}^{n} \log\left(1 + \frac{X_i}{\theta}\right) \quad (+1)$$

$$\Rightarrow \hat{\alpha} = \frac{n}{\sum_{i=1}^{n} \log\left(1 + \frac{X_i}{\theta}\right)} \quad (+1)$$

iii) Similar to tutorial

Fisher information $I(\alpha) = \frac{n}{\alpha^2}$ (+1)

$$CRLB = 1/I(\alpha) = \frac{\alpha^2}{n} \quad (+1)$$

iv) Mostly done in tut as is, except for first calculation

$$\mathbb{P}\left[\log\left(1 + \frac{X}{\theta}\right) \le y\right]$$

$$= \mathbb{P}\left[X \le \theta e^y - \theta\right]$$

$$= \int_0^{\theta e^y - \theta} \frac{\alpha}{\theta}\left(1 + \frac{x}{\theta}\right)^{-\alpha-1} dx$$

Let $x = \theta e^t - \theta \Rightarrow dx = \theta e^t dt$ (+1)

$$= \int_0^{y} \frac{\alpha}{\theta} e^{-(\alpha+1)t} \theta e^t \, dt$$

$$= \int_0^{y} \alpha e^{-\alpha t} dt \Rightarrow \log\left(1 + \frac{X}{\theta}\right) \sim Exp(\alpha)$$

sum of exponentials is Gamma by MGF

$$E[e^{tY}] = \prod_{i=1}^{n} E[e^{tX_i}] = \left(E[e^{tX}]\right)^n$$

$$= \left(\frac{\alpha}{\alpha - t}\right)^n \Rightarrow \text{Gamma}(n, \alpha) \quad (+1)$$

v) Similar to tutorial

$$\text{Bias} = E(\hat{\alpha}) - \alpha = n\,E(Y^{-1}) - \alpha$$

$$= n\frac{\alpha}{n-1} - \alpha$$

$$= \frac{\alpha}{n-1}$$

~~$\text{Bias} = |\alpha - E[\hat{\alpha}]| = |\alpha - n\,E[Y^{-1}]|$~~

~~$= |\alpha - n\frac{\alpha}{n-1}| = \frac{\alpha}{n-1}$~~ (+1)

$$\text{Variance} = E[\hat{\alpha}^2] - \left(E[\hat{\alpha}]\right)^2$$

$$= \frac{n^2\alpha^2}{(n-1)(n-2)} - \frac{n^2\alpha^2}{(n-1)^2}$$

$$= \frac{\alpha^2}{n}\left(\frac{n^3(n-1) - n^3(n-2)}{(n-1)^2(n-2)}\right)$$

$$= \frac{\alpha^2}{n}\left(\frac{n^3}{(n-1)^2(n-2)}\right) > \frac{\alpha^2}{n}$$

(+1) (+1)

b) Not seen before but similar calculation to tutorial

$$L(\theta; \underline{X}) = \frac{\alpha^n}{\theta^n}\prod_{i=1}^{n}\left(1 + \frac{x_i}{\theta}\right)^{-\alpha-1} \quad (+1)$$

No sufficient statistic exists (+1) because there's no way to write it as $h_1(\theta)\,h_2(\theta; t)$ where t is a statistic. (+1)

Question 3

Consider $\underline{X} = (X_1, \dots, X_n)$ with $X_i \overset{i.i.d.}{\sim} \text{Poi}(\lambda)$ for $i = \{1, \dots, n\}$ and $\lambda \in \mathbb{R}^+$ unknown.

(a) (bookwork, 5 marks)

We compute the likelihood ratio

$$\frac{f(\underline{x}; \lambda_0)}{f(\underline{x}; \lambda_1)} = \frac{\prod_{i=1}^n f(x_i; \lambda_0)}{\prod_{i=1}^n f(x_i; \lambda_1)} = \frac{\lambda_0^{\sum_{i=1}^n x_i} e^{-n\lambda_0}}{\prod_{i=1}^n x_i!} \cdot \frac{\prod_{i=1}^n x_i!}{\lambda_1^{\sum_{i=1}^n x_i} e^{-n\lambda_1}} \quad [1]$$

$$= \underbrace{\left(\frac{\lambda_0}{\lambda_1}\right)^{\sum_{i=1}^n x_i}}_{<1 \quad \forall \lambda_1 > \lambda_0} e^{\overbrace{-n(\lambda_0 - \lambda_1)}^{<0}} \quad \forall \lambda_0 < \lambda_1. \quad [1]$$

strictly monotonically decreasing as a function of the test statistic $(\sum_{i=1}^n x_i)$. [1]

Therefore there exists a UMP test for this problem with rejection region $R = \left\{ x : \sum_{i=1}^n x_i > k_\alpha \right\}$ [1]

with k_α adjusted to achieve a size α. [1]

b) (bookwork)

Because Poisson is closed under addition, [1] $(\sum_{i=1}^{n} X_i) \sim \text{Poi}(\lambda_0 n)$. To compute α, we use the approximation $(\sum_{i=1}^{n} X_i) \sim N(\lambda_0 n, \lambda_0 n)$ and find that [1] $\Pr[\Sigma X_i > k_\alpha] = \alpha = 0.025$ holds with [1]

$k_\alpha \simeq \lambda_0 n + 1.96\sqrt{\lambda_0 n} = 10196.$ [1]

[1]

c) (bookwork) In order to calculate the requested p-value we compute $\Pr\left[\sum_{i=1}^{n} X_i \geq \sum_{i=1}^{n} x_i = 10100\right]$ [2] $\simeq 1 - \Phi(z < \frac{10100-10000}{100})$ [1]

$= 1 - \Phi(z < 1)$

$= 1 - 0.8413$

$= 0.1587$

This is a large p-value, no significant evidence against the null hypothesis. [2]

$Z \sim N(0,1)$

[can include continuity correction]

Question 4 - F79MA (December 2023)

(a) (i)

Given that, $X \sim Geo(p)$, where $p \sim Beta(\alpha\ = 5,\ \beta = 7)$

Prior: $f(p) \propto p^4(1-p)^6$

[1 mark]

Likelihood:

$$L = \prod_{i=1}^{20} p(1-p)^{x_i - 1} = p^{20}(1-p)^{\sum x_i - 20} = p^{20}(1-p)^{520-20} = p^{20}(1-p)^{500}$$

$$where \sum x_i = n\bar{x} = 20 \times 26 = 520$$

[2 marks]

Posterior = Likelihood × Prior

$$= p^{20}(1-p)^{500}p^4(1-p)^6 = p^{24}(1-p)^{506} \sim Beta(25,\ 507)$$

[2 marks]

[SUBTOTAL: 5 marks]

(ii) Under quadratic loss the Bayesian estimator is the mean of the posterior distribution.

Mean of the Beta distribution,

$$Mean = \frac{\alpha}{\alpha + \beta} = \frac{25}{25 + 507} = 0.04699$$

[SUBTOTAL: 2 marks]

[TOTAL: 7 marks]

(b)

	1	2	3	$\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij}$	$\sum_{j=1}^{n}(X_{ij} - \bar{X}_i)^2$	$(\bar{X}_i - \bar{X})^2$
1	150	100	125	$\frac{150 + 100 + 125}{3}$ = 125	$(150 - 125)^2$ $+(100 - 125)^2$ $+(125 - 125)^2$ = 1250	$(125\text{-}105)^2 = 400$
2	100	110	120	110	200	25
3	80	90	70	80	200	625

$$\bar{X} = \frac{1}{N}\sum_{i=1}^{N}\bar{X}_i = \frac{125 + 110 + 80}{3} = 105$$

[1 mark]

$$E[s^2(\theta)] = \frac{1}{N}\sum_{i=1}^{N}\left\{\frac{1}{n-1}\sum_{j=1}^{n}(X_{ij} - \bar{X}_i)^2\right\} = \frac{1250 + 200 + 200}{3 \times (3-1)} = 275$$

[1 mark]

$$var[m(\theta)] = \frac{1}{N-1}\sum_{i=1}^{N}(\bar{X}_i - \bar{X})^2 - \frac{1}{Nn}\sum_{i=1}^{N}\left\{\frac{1}{n-1}\sum_{j=1}^{n}(X_{ij} - \bar{X}_i)^2\right\} = \frac{400 + 25 + 625}{3-1} - \frac{275}{3}$$
$$= 433.3333$$

[2 marks]

$$Z = \frac{n}{n + \frac{E[s^2(\theta)]}{var[m(\theta)]}} = \frac{3}{3 + \frac{275}{433.3333}} = 0.8254$$

[2 marks]

Credibility Premium for Risk 1 = $0.8254 \times 125 + (1 - 0.8254) \times 105 = 121.51$

Credibility Premium for Risk 2 = $0.8254 \times 110 + (1 - 0.8254) \times 105 = 109.13$

Credibility Premium for Risk 3 = $0.8254 \times 80 + (1 - 0.8254) \times 105 = 84.37$

[2 marks]

[TOTAL: 8 marks]